

Linear Regression - Subjective Questions

Assignment: BoomBikes Bike Sharing Demand Prediction

Part A - Assignment based Questions

Q1. From your analysis of the categorical variables in the dataset, what could you infer about their effect on the dependent variable? (3 marks)

I plotted box plots of **cnt** against each categorical variable during EDA. Here is what I noticed:

- bullet **season** - demand is highest in fall, then summer and winter, and lowest in spring. So spring is a strong negative driver.
- bullet **yr** - 2019 has clearly higher demand than 2018, which means bike sharing is becoming more popular year on year. So **yr** is a useful predictor and we should keep it (the assignment also warned us about this).
- bullet **mnth** - demand is high from May to October and drops in Dec to Feb. This matches the season pattern.
- bullet **weathersit** - demand decreases as weather worsens: clear > mist > light rain/snow. Bad weather is a strong negative driver.
- bullet **holiday** - holidays show slightly lower median demand than non-holidays.
- bullet **weekday** - the medians look similar for all weekdays, with Saturday showing a small positive lift.
- bullet **workingday** - working day vs non-working day medians are close, but working days have a slightly tighter spread.

Overall: the categorical variables - especially **season**, **weathersit**, **yr** and a few specific months - clearly affect the demand. So we should encode them as dummy variables before fitting the regression model.

Q2. Why is it important to use `drop_first=True` during dummy variable creation? (2 marks)

When a categorical variable with n levels is one-hot encoded, we get n dummy columns and they always sum to 1. That means one column can be perfectly predicted from the others. This is called the **dummy variable trap** and it makes the regression matrix singular - OLS cannot find a unique solution.

Setting **`drop_first=True`** drops one level so we end up with $n - 1$ dummies. The dropped level becomes the baseline (reference) and every other coefficient is interpreted as the effect of that level compared to the baseline. So this:

- bullet Removes the perfect multicollinearity (otherwise VIF will be infinite).
- bullet Reduces one feature per categorical variable, which makes the model simpler.
- bullet Keeps the coefficients easy to interpret - each one is a contrast against the baseline.

Q3. Looking at the pair plot among the numerical variables, which one has the highest correlation with the target variable? (1 mark)

temp has the highest correlation with **cnt** at around **0.63**. **atemp** (feels-like temperature) shows almost the same correlation, but it is nearly identical to temp (correlation ~0.99) so I dropped it before modelling to avoid multicollinearity.

Q4. How did you validate the assumptions of Linear Regression after building the model on the training set? (3 marks)

I checked the four classical assumptions one by one:

- bullet **Linearity** - I plotted residuals vs fitted values. The points were randomly scattered around 0 with no curve, so a linear relationship is reasonable.
- bullet **Normality of residuals** - I plotted a histogram with KDE and a Q-Q plot of the residuals. The histogram was roughly bell shaped and most points on the Q-Q plot fell on the 45 degree line, so the residuals are approximately normal.
- bullet **Homoscedasticity** (constant variance) - same residual vs fitted plot. There was no funnel or cone shape, so the variance of residuals is roughly constant.
- bullet **No multicollinearity** - I computed VIF for every predictor in the final model. All values are below 10 (highest is temp around 7.7) so multicollinearity is not a problem.
- bullet **Independence of errors** - the Durbin-Watson statistic from the OLS summary is around **2.06**, very close to 2, which confirms residuals are not auto-correlated.

Q5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of shared bikes? (2 marks)

Since I MinMax scaled all numerical features to [0, 1] before fitting, the size of each coefficient is directly comparable. Looking at the absolute coefficient values (all p-values are basically 0):

- bullet **temp** (+0.4039) - temperature is the strongest positive driver. Warmer days lead to noticeably more rentals.
- bullet **weathersit_light_rain_snow** (-0.2787) - bad weather is the strongest negative driver. Rain or snow days see a big drop in rentals.
- bullet **yr** (+0.2418) - 2019 has much higher demand than 2018, which means the platform is growing year on year.

Part B - General Subjective Questions

Q1. Explain the linear regression algorithm in detail. (4 marks)

Linear Regression is a supervised learning algorithm used to predict a continuous numerical value. The idea is simple - we model the target as a linear combination of one or more input features.

The two forms:

bullet **Simple linear regression:** $y = b_0 + b_1 * x + e$ (one predictor)

bullet **Multiple linear regression:** $y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_n * x_n + e$ (many predictors)

Here b_0 is the intercept, $b_1 \dots b_n$ are the slope coefficients and e is the error term.

How the model is trained:

bullet We use **Ordinary Least Squares (OLS)**, which finds the coefficients that minimise the sum of squared residuals: $\text{sum of } (y_{\text{actual}} - y_{\text{predicted}})^2$.

bullet OLS has a closed form solution: $b = (X'X)^{-1} * X'y$.

bullet For very large datasets people use gradient descent instead, but the idea is the same - minimise the squared error.

Key assumptions:

bullet **Linearity** - the relationship between features and target is linear in the parameters.

bullet **Independence** - the observations (and residuals) are independent of each other.

bullet **Homoscedasticity** - residuals have a constant variance.

bullet **Normality of residuals** - errors are approximately normally distributed. This matters mainly for p-values and confidence intervals.

bullet **No (or low) multicollinearity** - the predictors should not be strongly correlated with each other.

How we evaluate it: R-squared, Adjusted R-squared, MSE, RMSE and MAE are the common metrics.

Pros and cons: linear regression is simple, fast and very interpretable - we can clearly see which feature is doing what. The downside is that it cannot capture non-linear relationships on its own and it is sensitive to outliers and to features that are highly correlated with each other.

Q2. Explain Anscombe's quartet in detail. (3 marks)

Anscombe's quartet is a famous example created by the statistician Francis Anscombe in 1973. It is a set of four small datasets that look almost identical when you compute summary statistics, but look completely different when you plot them.

All four datasets share these summary statistics:

bullet Mean of x is about 9, mean of y is about 7.5

bullet Variance of x is 11, variance of y is about 4.12

bullet Correlation between x and y is about 0.816

bullet Same regression line: $y = 3.0 + 0.5 * x$

bullet Same R-squared, around 0.67

But when we plot them they look totally different:

- bullet **Dataset I** - a clean linear relationship with normal noise. Linear regression fits well.
- bullet **Dataset II** - a clear curved (non-linear) relationship. Linear regression cannot really capture this.
- bullet **Dataset III** - a perfect linear relationship except for one point that throws the line off (outlier).
- bullet **Dataset IV** - all x values are the same except one extreme point, and that single point decides the slope.

Why it matters: the quartet teaches us that we cannot trust summary statistics alone. Two datasets can have the exact same mean, variance, correlation and regression line but mean very different things. Always plot the data first - visualisation is essential before fitting any model, especially linear regression which is sensitive to outliers and non-linear patterns.

Q3. What is Pearson's R? (3 marks)

Pearson's correlation coefficient (R) is a number between -1 and +1 that tells us how strongly two continuous variables are **linearly** related and in which direction. The formula is:

$$R = \text{covariance}(X, Y) / (\text{std}(X) * \text{std}(Y))$$

How to read R:

- bullet R = +1 means perfect positive linear relationship.
- bullet R = -1 means perfect negative linear relationship.
- bullet R = 0 means no linear relationship at all.
- bullet Roughly: |R| below 0.3 is weak, 0.3 to 0.7 is moderate, above 0.7 is strong.
- Things to be careful about:
 - bullet R only catches **linear** patterns. Two variables can have a strong non-linear relationship and still have R close to 0.
 - bullet R is **sensitive to outliers** - a single extreme point can change it a lot.
 - bullet **Correlation is not causation** - high R only means the two variables move together, not that one is causing the other.
 - bullet Both variables should be continuous and ideally close to normally distributed for R to work well.

Q4. What is scaling? Why is scaling performed? What is the difference between normalised scaling and standardised scaling? (3 marks)

Scaling means transforming the numerical features so they are on the same range or scale. It changes the units, not the shape of the distribution.

Why we do it:

- bullet In linear models, scaling makes the coefficients **directly comparable**. If one feature is in thousands and another in tens, raw coefficients can be misleading.
- bullet Many ML algorithms - KNN, SVM, K-means, PCA, gradient descent based models, neural networks - rely on distances or step sizes, and they work much better when features are on a similar scale.

Regularised regressions (Ridge, Lasso) penalise coefficients fairly only when features are on the same scale.

Normalisation (Min-Max scaling):

$$x' = (x - \min(x)) / (\max(x) - \min(x))$$

Squeezes the values into a fixed range, usually [0, 1].

Keeps the original distribution shape.

Sensitive to outliers because min and max get pulled by extreme values.

Useful when the data does not follow a normal distribution and we want bounded values.

Standardisation (Z-score scaling):

$$x' = (x - \text{mean}) / \text{standard_deviation}$$

Centres the data around 0 with standard deviation 1.

Output is not bounded.

Handles outliers a little better than Min-Max.

Useful when the data is roughly normal or when the algorithm assumes standard-normal inputs.

Q5. You might have observed that sometimes the value of VIF is infinite. Why does this happen? (3 marks)

Variance Inflation Factor (VIF) for a feature X_j is calculated as:

$$\text{VIF} = 1 / (1 - R_j^2)$$

where R_j^2 is the R-squared value we get when we regress X_j on all the other features. VIF tells us how much the variance of that feature's coefficient is inflated because of multicollinearity.

VIF becomes **infinite when $R_j^2 = 1$** , meaning X_j can be predicted perfectly from the other features. The formula has $(1 - 1) = 0$ in the denominator. This happens when:

We forgot **drop_first=True** while creating dummies and kept all n levels of a categorical variable. The dummies sum to 1, so they are perfectly collinear.

We added a feature that is an exact transformation of others, like having both x and $2x + 5$ in the same model.

We accidentally added the same feature twice with two different names.

We have more features than rows of data ($n < p$), which forces perfect linear dependence.

Fix: find and remove the redundant feature - drop one dummy from a fully encoded categorical variable, or remove the duplicated/derived feature.

Q6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression. (3 marks)

A **Quantile-Quantile (Q-Q) plot** is a graph that compares the distribution of our sample data with a theoretical distribution, usually the standard normal. On the x-axis we plot the theoretical quantiles and on the y-axis we plot the sample quantiles.

How to read a Q-Q plot:

- bullet If most points lie on the 45 degree reference line, the sample distribution matches the theoretical one (so the data is approximately normal).
- bullet If both ends curve away from the line, the distribution has heavier or lighter tails than normal.
- bullet An S-shaped curve means the data is skewed.
- bullet If only the extreme points stray off the line, there are outliers in the tails.

Why it is important in linear regression:

- bullet One of the OLS assumptions is that the residuals are normally distributed. The Q-Q plot of residuals is the standard way to visually check this.
- bullet If residuals are not normal, then the p-values, t-tests and confidence intervals on the coefficients are not really trustworthy.
- bullet It also helps spot outliers and skewness, which might suggest we should transform a variable (log, square root etc.) or use a robust regression.
- bullet Together with the residuals vs fitted plot, the Q-Q plot gives a good picture of whether the regression assumptions hold.